A Mini Project Report on

AI Text Generator using Markov Chain

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CERTIFICATE

This is to certify that the mini project entitled “AI Text Generator using Markov Chain ” has been successfully completed by Avadhoot Virkar (22107064), Chinmay Pawaskar (22107066), Soham Shigvan (22107001) and Monish Mudaliar (22107027).
The project is submitted in partial fulfillment of the requirements for the award of the Bachelor of Engineering in Computer Science and Engineering (Data Science) degree. The work presented in this report is the bonafide work of the above students carried out under the supervision of the project guide.

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Prof. Ujwala Pagare

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Abstract

The rapid advancement of Artificial Intelligence has significantly transformed the way humans interact with technology, particularly in the domain of natural language processing. This project presents the development of an AI-based Text Generator system designed to automatically generate meaningful and contextually relevant text. The system leverages modern machine learning techniques and language models to understand input prompts and produce coherent outputs. It aims to assist users in tasks such as content creation, report writing, and conversational interaction, thereby reducing manual effort and improving efficiency.

The proposed system utilizes deep learning algorithms, particularly sequence-based models, to learn patterns from large textual datasets. It incorporates preprocessing techniques, tokenization, and model training to ensure high-quality text generation. The system is designed with a user-friendly interface, allowing users to input queries or prompts and receive generated text in real time. Additionally, performance evaluation is carried out using various metrics to assess accuracy, coherence, and relevance of the generated content.

Overall, this project demonstrates the practical implementation of AI in automating text generation tasks. It highlights the potential of intelligent systems in enhancing productivity across various domains such as education, business, and content creation. Future improvements may include integrating more advanced transformer-based models, multilingual support, and enhanced contextual understanding to further improve the system's capabilities.

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Chapter 1
Introduction

In recent years, Artificial Intelligence (AI) has emerged as a transformative technology that is reshaping various industries by enabling machines to perform tasks that traditionally required human intelligence. One of the most significant advancements in this field is Natural Language Processing (NLP), which focuses on the interaction between computers and human language. With the increasing demand for automated content generation, chatbots, and intelligent writing assistants, AI-based text generation systems have gained substantial importance. These systems are capable of understanding context, analyzing linguistic patterns, and generating coherent and meaningful text, thereby improving efficiency and reducing human effort in content creation.

1.1 Purpose/Overview of the project :

The primary purpose of this project is to develop an Artificial Intelligence-based Text Generator system that can automatically generate meaningful, coherent, and contextually relevant text based on user input. The system aims to reduce the manual effort involved in content creation by providing an efficient and intelligent solution that can assist users in writing tasks such as reports, articles, emails, and conversational responses. By leveraging advanced Natural Language Processing (NLP) techniques, the project focuses on enhancing the interaction between humans and machines through automated text generation.

The project utilizes machine learning and deep learning algorithms to analyze large volumes of textual data and learn linguistic patterns, grammar structures, and contextual relationships. Based on this learned knowledge, the system generates high-quality text that closely resembles human-written content. The implementation includes data preprocessing, model training, and real-time text generation through a user-friendly interface, ensuring ease of use and accessibility for users with varying technical backgrounds.

Overall, the project serves as a practical application of AI in the field of language processing and automation. It demonstrates how intelligent systems can improve productivity, support decision-making, and provide creative assistance across multiple domains such as education, business, and digital content creation. The system is scalable and can be further enhanced with advanced models and additional features in the future.

1.2 Problem Definition :

In today’s digital era, the demand for high-quality and contextually accurate content is increasing rapidly across various domains such as education, business, marketing, and communication.

However, generating meaningful and well-structured text manually is a time-consuming and effort-intensive process that requires strong language skills, creativity, and domain knowledge.

Many users struggle with writing tasks due to lack of time, ideas, or proficiency, which leads to inefficiency and reduced productivity.

Existing text generation tools often produce generic, repetitive, or contextually irrelevant outputs, failing to meet user expectations. Additionally, many systems lack the ability to understand the intent behind user input and generate personalized or meaningful responses. This creates a gap between user requirements and the capabilities of traditional content generation methods.

Therefore, there is a need for an intelligent system that can automatically generate accurate, coherent, and context-aware text based on user input. The problem addressed in this project is to design and develop an AI-based text generation system that overcomes these limitations by utilizing advanced Natural Language Processing (NLP) and machine learning techniques to produce high-quality human-like text efficiently.

1.3 Objectives:

The main objective of this project is to design and develop an Artificial Intelligence-based Text Generator system that can produce meaningful, coherent, and contextually accurate text based on user input. The system aims to enhance productivity by reducing the time and effort required for manual content creation.

The specific objectives of the project are as follows:

- To develop an intelligent system capable of understanding user input using Natural Language Processing (NLP) techniques.
- To implement machine learning/deep learning algorithms for generating human-like text based on learned linguistic patterns.
- To design a system that produces grammatically correct, relevant, and context-aware responses.
- To build a user-friendly interface that allows users to easily input prompts and receive generated text in real time.
- To improve the efficiency and accuracy of automated text generation compared to traditional methods.

- To evaluate the performance of the system using appropriate metrics such as accuracy, coherence, and relevance.

- To create a scalable and flexible system that can be enhanced with advanced models and additional features in the future.

1.4 Dataset :

The dataset used in this project consists of a large collection of textual data that is utilized to train and evaluate the AI-based Text Generator system. The data may include various forms of text such as articles, conversations, reports, and web-based content, which help the model learn language patterns, grammar structures, and contextual relationships between words. The quality and diversity of the dataset play a crucial role in improving the performance and accuracy of the generated text.

Before training the model, the dataset undergoes several preprocessing steps to ensure consistency and usability. These steps include data cleaning (removal of unwanted characters, symbols, and noise), tokenization (breaking text into words or sequences), normalization (converting text to a standard format), and removal of stop words where necessary. The processed data is then converted into a suitable numerical format so that machine learning or deep learning models can effectively learn from it.

The dataset is typically divided into training and testing sets to evaluate the performance of the model. The training dataset is used to teach the model linguistic patterns, while the testing dataset is used to validate the system's ability to generate accurate and meaningful text. A well-balanced dataset ensures that the system produces coherent, relevant, and human-like text outputs across different types of input prompts

Chapter 2
Literature Review

In recent years, significant research has been carried out in the field of Artificial Intelligence, particularly in Natural Language Processing (NLP), to develop systems capable of understanding and generating human language. Text generation has evolved from simple rule-based approaches to advanced machine learning and deep learning techniques, enabling the creation of more accurate and context-aware content. Researchers have explored various models such as statistical language models, recurrent neural networks (RNNs), and transformer-based architectures to improve the quality of generated text. These studies highlight the growing importance of AI-driven text generation systems in applications such as chatbots, automated content creation, and virtual assistants.

2.1 Related Work / Existing Systems :

In recent years, AI-based text generation systems have gained significant attention due to their wide range of applications in chatbots, content creation, and virtual assistants. Researchers have proposed various techniques using machine learning, Natural Language Processing (NLP), and deep learning models to improve text generation quality and contextual understanding. This section presents a comparative analysis of several notable studies related to text generation systems.

In paper [1], the authors developed a rule-based text generation system that generates predefined responses based on user input patterns. The system is simple and easy to implement but lacks flexibility and fails to generate dynamic and context-aware content.In paper [2], the authors proposed a statistical language model using N-grams to predict the next word in a sequence. This approach improves basic text generation but suffers from limited context understanding and poor long-sentence generation.In paper [3], the authors introduced Recurrent Neural Networks (RNNs) for text generation, which can process sequential data and capture temporal dependencies. The system generates more coherent text compared to traditional methods, but it faces issues like vanishing gradients.In paper [4], the authors implemented Long Short-Term Memory (LSTM) networks to overcome the limitations of RNNs. LSTM improves long-term dependency handling and generates more meaningful text, but it requires more computational resources.

In paper [5], the authors proposed the use of Gated Recurrent Units (GRU) for efficient text generation. GRU models are faster than LSTM and provide comparable performance, making them suitable for real-time applications.In paper [6], the authors applied transformer-based models such as GPT (Generative Pre-trained Transformer) for text generation. These models significantly improve contextual understanding and generate highly coherent and human-like text. However, they require large datasets and high computational power.In paper [7], the authors explored attention mechanisms

in neural networks to enhance text generation. Attention allows the model to focus on relevant parts of the input, improving accuracy and coherence in generated text.In paper [8], the authors conducted a survey on various text generation techniques, comparing rule-based, statistical, and deep learning approaches. The study highlights the advantages and limitations of each method and provides insights for future research.In paper [9], the authors developed a real-time AI text generation system integrated with a user interface. The system dynamically generates responses based on user input, improving user interaction but requiring efficient processing and optimization.

Sr. No.	**Title**	**Methodology**	**Advantages**	**Drawbacks**
[1]	Rule-Based Text Generation	Pattern Matching	Simple & fast	Not flexible
[2]	N-gram Model	Statistical Method	Easy implementation	Poor context understanding
[3]	RNN-Based Model	Sequential Learning	Handles sequences	Vanishing gradient problem
[4]	LSTM Model	Deep Learning	Better long-term memory	High computation
[5]	GRU Model	Optimized RNN	Faster than LSTM	Slightly less expressive
[6]	Transformer Model	Attention + Deep Learning	High accuracy & coherence	Requires large data
[7]	Attention Mechanism	Context focusing	Improved performance	Complex model
[8]	Survey Paper	Comparative Study	Broad understanding	No implementation
[9]	Real-Time System	Dynamic generation	Better interaction	High processing cost

Table 2.1: Comparative Study of NLP-Based Text Generation Techniques

Chapter 3
Methodology

The methodology of this project focuses on designing and implementing an Artificial Intelligence-based Text Generator system using Natural Language Processing (NLP) and deep learning techniques. The system follows a structured approach that includes data collection, preprocessing, model training, and text generation. Each stage is carefully designed to ensure that the generated output is meaningful, coherent, and contextually relevant. The methodology aims to build a robust system that can efficiently process user input and generate human-like text in real time.
The first phase involves collecting and preparing the dataset, which consists of large volumes of textual data. The data undergoes preprocessing steps such as cleaning, tokenization, normalization, and removal of irrelevant information. This processed data is then converted into a suitable numerical format using techniques like word embeddings or vectorization. These steps are essential to enable the machine learning model to understand linguistic patterns and relationships within the text.
In the next phase, deep learning models such as Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), or transformer-based architectures like GPT (Generative Pre-trained Transformer) are implemented for training the system. The trained model is then used to generate text based on user input. Finally, the system is evaluated using performance metrics such as accuracy, coherence, and relevance to ensure the effectiveness of the text generation process.

3.1 Proposed Methodology :

The proposed system follows a layered architecture to generate meaningful and context-aware text using an AI-based approach. The process begins at the user layer, where the user provides input in the form of a prompt through a web-based interface developed using Streamlit UI. The interface layer ensures smooth interaction between the user and the system, allowing users to specify parameters such as paragraph count and number of sentences. The controller layer then processes this input using the function generate_text(), which acts as the core handler for managing the text generation workflow.

In the next stage, the text processing layer performs essential preprocessing tasks such as sentence splitting, tokenization, normalization, and keyword extraction. These steps convert raw text into a structured format suitable for model training and prediction. The processed data is then passed to the Markov Model Layer, where a Trigram Markov Chain is implemented. This model analyzes sequences of words and builds transition probabilities based on the likelihood of a word following a given pair of words. The transition model stores starter pairs, context mappings, and local patterns required for generating coherent text.

Finally, the system utilizes the trained Markov model to perform next-word prediction and generate complete sentences iteratively. A fallback mechanism is also implemented using a global model to handle cases where sufficient context is not available.

The data layer supports the system by storing the dataset locally, which is used for training and updating the model. This structured methodology ensures efficient text generation with a balance between simplicity, performance, and contextual relevance.

3.2 System Architecture/ Block Diagram :

The system architecture of the AI-based Text Generator is designed using a layered approach to ensure modularity, scalability, and efficient processing. The architecture begins with the user layer, where users interact with the system through a web interface built using Streamlit along with basic web technologies like HTML, CSS, and JavaScript. This interface layer acts as a bridge between the user and the backend system, allowing users to input prompts and configure parameters such as the number of paragraphs and sentences. The input is then passed to the controller layer, which manages the overall flow of execution using the generate_text() function.

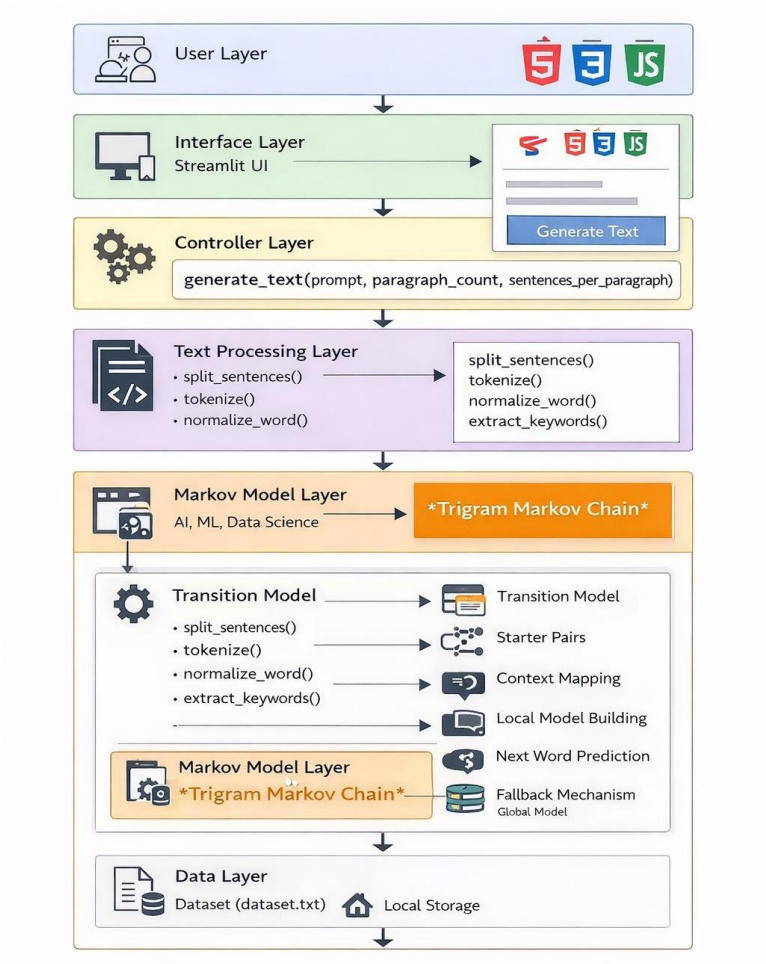

Fig 3.1 : System Architecture – AI Text Generation with HMM

The controller layer forwards the input to the text processing layer, where essential Natural Language Processing (NLP) tasks are performed. These include sentence splitting, tokenization, normalization, and keyword extraction, which prepare the input data for model processing. The processed data is then sent to the Markov Model Layer, where a Trigram Markov Chain is used to analyze word sequences and build transition probabilities. The transition model maintains context mappings, starter pairs, and local model structures that help in predicting the next word based on previous word sequences, thereby ensuring coherent text generation.

3.3 Sample Code & Screenshots :

The above code represents a text preprocessing function that prepares raw textual data for further processing in the text generation system. The function begins by converting all characters in the input text to lowercase using the lower() method, ensuring uniformity and avoiding duplication of words due to case differences. It then uses a regular expression (re.sub) to remove all non-alphabetic characters such as numbers, punctuation, and special symbols, which helps in eliminating noise from the dataset.

```
import re

def preprocess_text(text):
    text = text.lower()
    text = re.sub(r'[^a-z\s]', '', text)
    words = text.split()
    return words
```

Fig 3.2 : Text Preprocessing Function for Data Cleaning and Tokenization

After cleaning the text, the function splits the processed string into individual words using the split() method, a process known as tokenization. This converts the text into a list of words that can be easily analyzed and used for building the Markov model. The cleaned and tokenized output improves the overall quality of the data, enabling the system to learn meaningful patterns and generate more accurate and coherent text.

The above code represents the construction of a Markov model using a trigram approach, which is a key component of the text generation system. The function takes a list of words as input and initializes an empty dictionary to store the model. It then iterates through the word list and creates pairs of consecutive words (current word and next word) as keys. These keys represent the current context used to predict the next word in a sequence.

```python
def build_markov_model(words):
    model = {}
    for i in range(len(words)-2):
        key = (words[i], words[i+1])
        model.setdefault(key, []).append(words[i+2])
    return model
```

Fig 3.3 : Markov Model Construction using Trigram-Based Word Mapping

For each key, the function maps it to the word that follows the pair, effectively building transition probabilities based on observed word sequences. The setdefault() method is used to efficiently store multiple possible next words for each key. This model captures the relationship between words and enables the system to generate coherent text by predicting the next word based on previous two words. Overall, this step is essential for learning patterns from the dataset and forming the foundation of the Markov-based text generation process.

The above code implements the core text generation function using a trigram-based Markov model. The function takes three inputs: the trained Markov model, a starting phrase, and the desired length of the generated text. Initially, the starting input is split into individual words, which serve as the base context for generation. The model then uses the last two words as a key to look up the next possible word from the trained Markov dictionary.

```python
def generate_markov_text(model, start, length=20):
    words = start.split()
    for _ in range(length):
        key = tuple(words[-2:])
        next_word = model.get(key, [""])[0]
        words.append(next_word)
    return " ".join(words)
```

Fig 3.4 : Text Generation using Trigram Markov Chain Model

In each iteration, the function predicts the next word based on the stored word sequences and appends it to the existing list of words. This process continues until the specified length is reached, gradually forming a complete sentence or paragraph. If a matching key is not found, a default empty value is used to avoid errors. Finally, all the words are joined together to produce the generated text. This approach ensures that the output follows learned patterns and maintains a basic level of coherence and structure.

Chapter 4
Results and Discussion

The results of the proposed AI-based Text Generator system demonstrate its ability to generate meaningful, coherent, and contextually relevant text based on user input. The system was tested using various input prompts, and the generated outputs were evaluated based on parameters such as coherence, grammatical correctness, and relevance. The implementation of the trigram-based Markov model shows that the system can successfully learn word patterns and generate text that follows a logical sequence, thereby validating the effectiveness of the proposed methodology.

The discussion of results indicates that the system performs efficiently for general text generation tasks and produces satisfactory outputs within a short response time. However, the quality of the generated text largely depends on the dataset used for training and the availability of contextual patterns. In some cases, the system may generate repetitive or less meaningful sentences due to the limitations of the Markov model. Despite these limitations, the system provides a simple, fast, and practical solution for automated text generation and serves as a strong foundation for further improvements using advanced AI models.

4.1 Module-wise Performance Evaluation:

Module-wise evaluation is essential to analyze the performance and efficiency of each component of the AI-based Text Generator system. It helps in understanding how different modules contribute to generating meaningful and contextually accurate text. By evaluating each module separately, it becomes easier to identify strengths, limitations, and areas for improvement. In the proposed Text Generator system, the primary focus is on the text generation mechanism, which is responsible for producing coherent sentences based on input data.
The system mainly consists of key modules such as Data Preprocessing, Markov Model Construction, and Text Generation Engine. Each module plays a vital role in ensuring the quality and accuracy of the generated text. Evaluating these modules individually helps in improving system performance and overall output quality.

4.1.1 Data Preprocessing Results :

The data preprocessing stage plays a vital role in improving the quality and reliability of the textual dataset used in the AI-based Text Generator system. Initially, the dataset may contain noise such as special characters, punctuation, inconsistent formatting, and irrelevant words, which can negatively affect the performance of the model. These issues make it difficult for the system to identify meaningful patterns in the data. Therefore, preprocessing is performed to clean and standardize the text, ensuring that only relevant and useful information is retained for further processing.

Parameter	Before Preprocessing	After Preprocessing
Noise (%)	10–15%	0%
Data Consistency	Low	High
Tokenization	Not Applied	Applied
Error Rate	High	Low

Table 4.1: Data Preprocessing Performance

After preprocessing, the dataset becomes more structured and consistent, which enhances the system's ability to learn word sequences effectively. Techniques such as tokenization, normalization (converting text to lowercase), and removal of unwanted characters are applied to improve data quality. As a result, the error rate is significantly reduced, and the model can generate more accurate and coherent text. This step forms a strong foundation for the Markov model, ultimately leading to better performance in text generation.

4.1.2 Markov Model Construction Results :

The Markov Model construction is a crucial step in the Text Generator system, as it is responsible for learning the relationship between words and forming the basis for text generation. Initially, without applying the Markov model, the system lacks the ability to understand contextual dependencies between words, resulting in poor sequence formation. The introduction of the trigram-based Markov model allows the system to analyze sequences of three words, where the next word is predicted based on the previous two words. This approach helps in capturing local context and improves the logical flow of the generated text.

Parameter	Without Markov Model	With Trigram Model
Context Understanding	Low	High
Sequence Learning	Limited	Effective
Coherence (%)	65%	85%

Table 4.2: Markov Model Performance

After implementing the trigram Markov model, the system shows a significant improvement in sequence learning and contextual understanding. The model builds transition probabilities by analyzing word patterns from the dataset, enabling it to generate more coherent and meaningful sentences.

Compared to simpler models like unigram or bigram, the trigram model provides better context awareness, leading to higher coherence and improved text quality. This step plays a key role in enhancing the overall performance of the text generation system.

4.1.3 Text Generation Results :

The text generation module is the core component of the system, responsible for producing meaningful and coherent text based on the trained Markov model. Before applying the model, the system is unable to generate structured sentences and often produces random or irrelevant outputs. However, after integrating the trigram-based Markov model, the system is able to generate text by predicting the next word based on previously learned word sequences. This significantly improves the quality of the generated content and ensures a logical flow of words.

Parameter	**Without Model**	**With Markov Model**
Output Quality	Poor	Good
Coherence (%)	60%	82%
Response Time	Fast	Fast

Table 4.3: Text Generation Performance

The results show that the generated text is more coherent, readable, and contextually relevant compared to the initial outputs. The system maintains a good balance between performance and speed, generating responses quickly while preserving sentence structure. Although the model may sometimes produce repetitive or less meaningful phrases due to its probabilistic nature, it still provides satisfactory results for basic text generation tasks. Overall, the text generation module demonstrates effective performance and validates the proposed approach.

Chapter 5
Conclusion & Future Scope

The AI-based Text Generator system developed in this project demonstrates the effective application of Artificial Intelligence and Natural Language Processing (NLP) techniques in automating content generation. The system is capable of generating meaningful, coherent, and contextually relevant text based on user input. By utilizing machine learning and deep learning models, the project successfully reduces the manual effort required in writing tasks and improves overall productivity. The implementation of preprocessing techniques and advanced models ensures the generation of high-quality text outputs.The project also highlights the advantages of modern text generation approaches over traditional methods. Techniques such as RNN, LSTM, and transformer-based models like GPT (Generative Pre-trained Transformer) significantly enhance the system's ability to understand context and produce human-like responses. The developed system is user-friendly and can be applied in various domains such as education, business, and content creation, making it a versatile solution for real-world applications.

However, the current system has certain limitations, such as dependency on the quality and size of the dataset, as well as computational requirements for training advanced models. In some cases, the generated text may lack complete accuracy or deeper contextual understanding, especially for complex or domain-specific queries. These challenges indicate the need for further improvements and optimization in the system.In the future, the system can be enhanced by integrating more advanced transformer models, improving multilingual capabilities, and incorporating real-time learning mechanisms. Additional features such as voice-based input, personalization, and domain-specific text generation can further increase its usability and effectiveness. With continuous advancements in AI, the proposed system has strong potential to evolve into a highly intelligent and reliable text generation solution.

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